

Binned K -Propagation for Coordinate-Spiked ReLU Networks

A hidden-Markov model over the spike coordinate with conditional bulk cumulant propagation

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Abstract

We describe BINNED-KPROP, a mechanistic predictor of $\mathbb{E}[f(X)]$, $X \sim \mathcal{N}(0, I_n)$, for a ReLU MLP f whose hidden weight matrices carry a single-coordinate rank-one spike $M = W + e_1 e_1^\top$. Ordinary total-order cumulant propagation (*kprop*) truncated at order K fails here: the spike coordinate retains $O(1)$ cumulants at *every* order, so no fixed truncation captures it. The algorithm instead splits each layer vector into the scalar spike coordinate A and the bulk $B \perp e_1$, represents A non-parametrically by a discrete law over m bins evolving as a hidden-Markov model between layers, and propagates B by ordinary $K = 2$ cumulant propagation *conditionally on each bin*. Every update is in closed form (Gaussian and truncated-Gaussian integrals). Bin edges and representatives are chosen to minimize the Wasserstein-2 distance to the exact per-layer law of A , which under the $K = 2$ closure is a Gaussian mixture; the W_2 -optimal quantizer is computed by Lloyd–Max iteration with closed-form mixture centroids. Empirically the predictor recovers the budget rate $\text{MSE} \sim n^{-2}$ in the width n , while the naive one-bin closure only attains n^{-1} .

1 Setting and motivation

Model. Let f be a depth- L ReLU MLP with square hidden layers of width n ,

$$X^{(0)} \sim \mathcal{N}(0, I_n), \quad Z^{(\ell)} = M^{(\ell)} X^{(\ell-1)} + b^{(\ell)}, \quad X^{(\ell)} = \rho(Z^{(\ell)}),$$

followed by a linear readout. Each hidden matrix carries a coordinate spike on the axis $e = e_1$:

$$M = W + e_1 e_1^\top, \quad W_{ij} \sim \mathcal{N}(0, 1/n).$$

The target is the output mean $\mathbb{E}[f(X)]$.

Spike/bulk split. Every layer vector is decomposed along the spike axis,

$$X = A e + B, \quad A = e^\top X \in \mathbb{R}, \quad B = \Pi X \in e^\perp, \quad \Pi = I - e e^\top, \quad (1)$$

where A is the scalar *spike coordinate* and B the d -dimensional *bulk*, $d = n - 1$. Relative to $\mathbb{R}^n = \mathbb{R}e \oplus e^\perp$ a hidden matrix block-decomposes as

$$M = \begin{pmatrix} \gamma & r^\top \\ u & V \end{pmatrix}, \quad \gamma = M_{00}, \quad r = M_{0,1:}, \quad u = M_{1:,0}, \quad V = M_{1:,1:},$$

so one linear layer acts as

$$\boxed{A^+ = \gamma A + r^\top B, \quad B^+ = u A + V B.} \quad (2)$$

Why bin the spike. For a *flat* spike $\frac{1}{n}\mathbf{1}\mathbf{1}^\top$, the spike direction picks up a flat-loop $1/n$ discount in the diagrammatics and ordinary total-order kprop is accurate. A *coordinate* spike has no such discount: the map (2) preserves an $O(1)$ residue of coordinate 0 ($\gamma = W_{00} + 1 \approx 1$), so the cumulants of A are $O(1)$ at **every** order and cannot be truncated at any fixed K . The remedy is to stop parameterizing A by cumulants altogether: represent its law *non-parametrically* as a discrete distribution over m bins — a hidden-Markov model over the spike coordinate, with a scalar transition kernel between layers — and run ordinary $K = 2$ cumulant propagation on the bulk B *conditional on each bin*.

2 State

The joint law of a layer vector X is approximated by the mixture

$$\hat{\mathcal{L}}(X) = \sum_{\alpha=1}^m p_\alpha \delta_{v_\alpha}(dA) \otimes \nu_\alpha(dB), \quad \nu_\alpha \approx \mathcal{N}(\mu_\alpha, \Sigma_\alpha), \quad (3)$$

stored as four arrays:

symbol	shape	meaning
p_α	$(m,)$	$P(A \in \text{bin } \alpha)$; nonnegative, sums to 1
v_α	$(m,)$	representative spike value $\approx \mathbb{E}[A \mid \text{bin } \alpha]$
μ_α	(m, d)	conditional bulk mean $\mathbb{E}[B \mid \text{bin } \alpha]$
Σ_α	(m, d, d)	conditional bulk covariance $\text{Cov}[B \mid \text{bin } \alpha]$

Coordinate 0 never appears inside μ or Σ ; the spike lives only in (p, v) . Within a bin, A is collapsed to the single point v_α — this is the one approximation binning introduces in the spike direction, and Section 7 chooses the bins to minimize it.

3 Initialization

With $X^{(0)} \sim \mathcal{N}(0, I_n)$ we have $A^{(0)} \sim \mathcal{N}(0, 1)$ independent of $B^{(0)} \sim \mathcal{N}(0, I_d)$. Given pre-activation edges $z_0 = -\infty < z_1 < \dots < z_m = +\infty$ defining bins $[z_{\alpha-1}, z_\alpha)$,

$$p_\alpha = \Phi(z_\alpha) - \Phi(z_{\alpha-1}), \quad v_\alpha = \mathbb{E}[A^{(0)} \mid \text{bin } \alpha] = \frac{\varphi(z_{\alpha-1}) - \varphi(z_\alpha)}{\Phi(z_\alpha) - \Phi(z_{\alpha-1})}, \quad \mu_\alpha = 0, \quad \Sigma_\alpha = I_d, \quad (4)$$

where φ, Φ are the standard-normal pdf and cdf. The truncated-normal mean v_α is already the W_2 -optimal representative for the cell (Section 7).

4 Linear step

Fix an old bin α : inside it, $A = v_\alpha$ is a point and $B \sim \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$. Push both through (2).

4.1 The joint Gaussian (Y, C)

The new scalar spike pre-activation and new bulk vector are one affine map of the Gaussian B ,

$$\begin{pmatrix} Y_\alpha \\ C_\alpha \end{pmatrix} = \begin{pmatrix} \gamma \\ u \end{pmatrix} v_\alpha + \begin{pmatrix} r^\top \\ V \end{pmatrix} B,$$

hence jointly Gaussian, with mean and covariance read off directly:

$$\begin{pmatrix} Y_\alpha \\ C_\alpha \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} m_Y \\ m_C \end{pmatrix} = \begin{pmatrix} \gamma \\ u \end{pmatrix} v_\alpha + \begin{pmatrix} r^\top \\ V \end{pmatrix} \mu_\alpha, \begin{pmatrix} s_Y^2 & g^\top \\ g & \Sigma_C \end{pmatrix} = \begin{pmatrix} r^\top \\ V \end{pmatrix} \Sigma_\alpha \begin{pmatrix} r^\top \\ V \end{pmatrix}^\top\right). \quad (5)$$

4.2 Transition kernel

For each new bin $I_\beta = [\ell_\beta, h_\beta)$, with standardized limits $a = (\ell_\beta - m_Y)/s_Y$ and $b = (h_\beta - m_Y)/s_Y$,

$$Q_{\beta\alpha} = P(Y_\alpha \in I_\beta) = \Phi(b) - \Phi(a).$$

New bin masses follow by marginalization, and Bayes gives the posterior of the old bin given the new:

$$p_\beta^+ = \sum_\alpha p_\alpha Q_{\beta\alpha} \quad (p^+ = Qp), \quad \eta_{\alpha|\beta} = \frac{p_\alpha Q_{\beta\alpha}}{p_\beta^+}. \quad (6)$$

This is exactly one step of a hidden-Markov filter with emission “which bin did Y land in.”

4.3 Conditioning the bulk on $Y \in I_\beta$

The truncated-normal moments of Y on I_β (writing $Z = Q_{\beta\alpha}$) are

$$\tau_1 = \mathbb{E}[Y - m_Y \mid I_\beta] = s_Y \frac{\varphi(a) - \varphi(b)}{Z}, \quad \tau_2 = \mathbb{E}[(Y - m_Y)^2 \mid I_\beta] = s_Y^2 \left[1 + \frac{a\varphi(a) - b\varphi(b)}{Z}\right],$$

so $\text{Var}(Y \mid I_\beta) = \tau_2 - \tau_1^2$. Because (Y, C) is jointly Gaussian, the regression of C on Y is exact:

$$C_\alpha = m_C + \frac{g}{s_Y^2}(Y_\alpha - m_Y) + C_\perp, \quad C_\perp \perp Y_\alpha, \quad \text{Cov}(C_\perp) = \Sigma_C - \frac{gg^\top}{s_Y^2}.$$

Taking conditional mean and covariance over $Y \in I_\beta$ (only the Y -dependent part feels the truncation),

$$\boxed{\mu_{\alpha \rightarrow \beta} = m_C + \frac{g}{s_Y^2} \tau_1, \quad \Sigma_{\alpha \rightarrow \beta} = \underbrace{\Sigma_C - \frac{gg^\top}{s_Y^2}}_{\text{cov. given exact } Y} + \underbrace{\frac{gg^\top}{s_Y^4} (\tau_2 - \tau_1^2)}_{\text{re-add } Y\text{'s in-bin spread}}.} \quad (7)$$

Degenerate branch $s_Y^2 \approx 0$: Y_α is the constant m_Y , so it lands wholly in the bin containing m_Y , with $\mu_{\alpha \rightarrow \beta} = m_C$ and $\Sigma_{\alpha \rightarrow \beta} = \Sigma_C$ (no division by s_Y^2).

4.4 Mixture over old bins

Several old bins feed each new bin β ; they combine with the posterior weights (6) by the laws of total mean and total covariance:

$$v_\beta^+ = \sum_{\alpha} \eta_{\alpha|\beta} \mathbb{E}[Y_\alpha | I_\beta] = \sum_{\alpha} \eta_{\alpha|\beta} (m_{Y,\alpha} + \tau_{1,\alpha\beta}), \quad (8)$$

$$\mu_\beta^+ = \sum_{\alpha} \eta_{\alpha|\beta} \mu_{\alpha \rightarrow \beta}, \quad \Sigma_\beta^+ = \sum_{\alpha} \eta_{\alpha|\beta} \left[\Sigma_{\alpha \rightarrow \beta} + (\mu_{\alpha \rightarrow \beta} - \mu_\beta^+) (\mu_{\alpha \rightarrow \beta} - \mu_\beta^+)^{\top} \right]. \quad (9)$$

The outer-product term is the between-component spread. Note v_β^+ is the *dynamic* conditional mean $\mathbb{E}[A^+ | A^+ \in I_\beta]$, not a fixed bin center. Finally p^+ is renormalized (logging the tail mass lost) and Σ^+ is symmetrized and PSD-clipped (logging any clip); both corrections are $\sim 10^{-16}$ in practice.

Cost. The only $O(n^3)$ work is $\Sigma_C = V \Sigma_\alpha V^{\top}$, once per old bin; everything else is $O(m^2 d + m d^2)$ per layer. The implementation never forms an (m, m, d, d) tensor: it stores the β -independent part $\Sigma_C - gg^{\top}/s_Y^2$ once per α and re-adds the rank-one $gg^{\top}$ term per new bin. Total: $O(m n^3)$ per linear layer.

5 ReLU step

ReLU is coordinatewise, so it respects the split (1): $\rho(X) = \rho(A) e + \rho(B)$.

5.1 Bulk, inside each bin

With $B \sim \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$, $\sigma_i^2 = \Sigma_{\alpha,ii}$, $z_i = \mu_i/\sigma_i$, the exact rectified-Gaussian marginal moments are

$$\mathbb{E}[\rho(B_i)] = \mu_i \Phi(z_i) + \sigma_i \varphi(z_i), \quad \mathbb{E}[\rho(B_i)^2] = (\mu_i^2 + \sigma_i^2) \Phi(z_i) + \mu_i \sigma_i \varphi(z_i),$$

giving $\tilde{\mu}_{\alpha,i} = \mathbb{E}[\rho(B_i)]$ and $\tilde{\Sigma}_{\alpha,ii} = \mathbb{E}[\rho(B_i)^2] - \tilde{\mu}_{\alpha,i}^2$. Off-diagonal entries (default backend "exact") use the exact bivariate-Gaussian ReLU covariance $\text{Cov}(\rho(B_i), \rho(B_j))$ via Owen's T ; the cheaper "gain" backend uses the leading-order $\tilde{\Sigma}_{ij} = \Phi(z_i) \Phi(z_j) \Sigma_{ij}$; "kprop" delegates to the ordinary harmonic-kprop ReLU (which also enables per-bin propagation at general $k_{\max} > 2$).

5.2 Spike and bin merge

ReLU output is nonnegative, so the representatives map $\tilde{v}_\alpha = \max(v_\alpha, 0)$ onto a *nonnegative* post-activation grid with a dedicated zero bin. Old bins landing in the same post bin β , $S_\beta = \{\alpha : \text{bin}(\tilde{v}_\alpha) = \beta\}$, merge by mixture moments with $\eta_{\alpha|\beta} = p_\alpha/p_\beta^+$:

$$p_\beta^+ = \sum_{\alpha \in S_\beta} p_\alpha, \quad v_\beta^+ = \sum_{\alpha \in S_\beta} \eta_{\alpha|\beta} \tilde{v}_\alpha, \quad \mu_\beta^+ = \sum_{\alpha \in S_\beta} \eta_{\alpha|\beta} \tilde{\mu}_\alpha, \quad \Sigma_\beta^+ = \sum_{\alpha \in S_\beta} \eta_{\alpha|\beta} [\tilde{\Sigma}_\alpha + (\tilde{\mu}_\alpha - \mu_\beta^+) (\tilde{\mu}_\alpha - \mu_\beta^+)^{\top}]. \quad (10)$$

Typically all bins with $v_\alpha \leq 0$ collapse into the zero bin — that is the post-ReLU mass atom at 0.

Algorithm 1: BINNED-KPROP ($K = 2$, m spike bins)

1. Build pre-activation edges (with negative bins) and post-ReLU edges (nonnegative, zero bin).
 2. $(p, v, \mu, \Sigma) \leftarrow \text{GAUSSIANINITIALSTATE}(d, \text{edges})$: truncated-normal p_α, v_α ; $\mu = 0$, $\Sigma = I_d$.
 3. For each hidden layer with matrix $M = \begin{pmatrix} \gamma & r^\top \\ u & V \end{pmatrix}$ and bias:
 - (a) **Linear step:** for each old bin α compute $(m_Y, s_Y^2, m_C, \Sigma_C, g)$ by (5);
 - (b) transition kernel $Q_{\beta\alpha}$, masses $p^+ = Qp$, posteriors $\eta_{\alpha|\beta}$ (eq. (6))
 - (c) condition the bulk on $Y \in I_\beta$: $(\mu_{\alpha \rightarrow \beta}, \Sigma_{\alpha \rightarrow \beta})$ (eq. (7))
 - (d) merge over old bins: (v^+, μ^+, Σ^+) (eqs. (8)–(9))
 - (e) renormalize p^+ ; symmetrize/PSD-clip Σ^+ ;
 - (f) **ReLU step:** per bin, exact rectified-Gaussian bulk update $(\tilde{\mu}_\alpha, \tilde{\Sigma}_\alpha)$; $\tilde{v}_\alpha = \max(v_\alpha, 0)$;
 - (g) re-bin $\tilde{v}$ on the post grid and merge coincident bins by mixture moments.
 4. Return $W_{\text{ro}}(\bar{A}e + \bar{B}) + b_{\text{ro}}$ with $\bar{A} = \sum_\alpha p_\alpha v_\alpha$, $\bar{B} = \sum_\alpha p_\alpha \mu_\alpha$.
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6 Readout and recovered moments

A full network alternates `LINEAR_STEP` (with pre-activation edges, which must include negative bins since $A^+ = \gamma A + r^\top B$ can be negative even when the incoming $A \geq 0$) and `RELU_STEP` (with nonnegative post edges). The readout is linear, so after the last hidden ReLU the full mean is reconstructed as

$$\mathbb{E}[X] = \bar{A}e + \bar{B}, \quad \bar{A} = \sum_\alpha p_\alpha v_\alpha, \quad \bar{B} = \sum_\alpha p_\alpha \mu_\alpha, \quad (11)$$

and the output mean is $W_{\text{ro}}\mathbb{E}[X] + b_{\text{ro}}$. The full covariance, when needed, follows by the law of total covariance over the finite spike law.

7 Bin placement by Wasserstein-2 quantization

7.1 The right error to minimize

Binning replaces the continuous law of A by the m -atom law $\hat{A} = \sum_\alpha p_\alpha \delta_{v_\alpha}$. Within a cell every value of A is reported as v_α , so the squared error introduced in the spike direction is

$$\sum_\alpha \int_{\text{cell } \alpha} (a - v_\alpha)^2 f(a) da = \sum_\alpha p_\alpha \text{Var}(A \mid \text{cell } \alpha) = W_2^2(\mathcal{L}(A), \hat{\mathcal{L}}(A)), \quad (12)$$

the squared Wasserstein-2 distance between the true continuous A and its binned version (the nearest-representative coupling is optimal). “Choose the bins well” therefore *means* “minimize W_2 to the expected continuous distribution of A ”: (12) is exactly the in-bin variance the linear step’s representative collapse throws away.

7.2 Lloyd–Max optimality conditions

Minimizing (12) over both partition and representatives yields the two stationarity conditions of optimal scalar quantization: (i) representatives are cell centroids, $v_\alpha = \mathbb{E}[A \mid \text{cell } \alpha]$; (ii) edges are midpoints of adjacent representatives, $z_\alpha = \frac{1}{2}(v_\alpha + v_{\alpha+1})$. Condition (i) is what the propagation already uses everywhere (all the conditional means of Sections 3–5). Condition (ii) is what a fixed equal-mass quantile grid does *not* satisfy. Alternating (i) and (ii) is Lloyd’s algorithm; for log-concave densities (Gaussian, rectified Gaussian) it converges to the global optimum.

7.3 The exact per-layer law of A is a Gaussian mixture

Under the $K = 2$ closure the expected continuous law of the next spike coordinate is available in closed form, and that is what gets quantized:

- **Pre-activation grid.** A^+ is exactly a Gaussian mixture, one component per current bin:

$$\mathcal{L}(A^+) = \sum_{\alpha} p_{\alpha} \mathcal{N}(m_{Y,\alpha}, s_{Y,\alpha}^2), \quad m_{Y,\alpha} = \gamma v_{\alpha} + r^{\top} \mu_{\alpha}, \quad s_{Y,\alpha}^2 = r^{\top} \Sigma_{\alpha} r.$$

The cell centroid of a mixture is closed form (no quadrature): with $\alpha_k = (a - m_k)/s_k$, $\beta_k = (b - m_k)/s_k$,

$$\mathbb{E}[A^+ \mid [a, b]] = \frac{\sum_k w_k [m_k(\Phi(\beta_k) - \Phi(\alpha_k)) + s_k(\varphi(\alpha_k) - \varphi(\beta_k))]}{\sum_k w_k (\Phi(\beta_k) - \Phi(\alpha_k))}.$$

This coincides with the linear step’s representative v_{β}^+ in (8), so the propagated per-bin value is the *exact mixture centroid*, never a single-Gaussian approximation.

- **Post-ReLU grid.** $A_{\text{post}} = \max(A_{\text{pre}}, 0)$ is the rectified mixture: an atom at 0 of mass $\sum_k w_k \Phi(-m_k/s_k)$ plus a truncated positive tail. Its W_2 -optimal quantizer pins one representative on the 0-atom and Lloyd–Max points on the tail.

Two grid builders exist: `lloyd_max_edges` (single moment-matched Gaussian, cheap) and `lloyd_max_edges_mixture` (the true mixture, closed-form centroids each iteration). On a 3-component test the mixture version cut W_2^2 to the true law by 14–28% over the moment-matched one.

7.4 W_2 -optimal versus equal-mass quantiles

Equal-mass quantiles place point density $\propto f$; the squared-error optimum places density $\propto f^{1/3}$ (Panter–Dite high-rate quantization), i.e. more resolution in the tails — exactly where the amplified spike coordinate carries the mass that ReLU treats asymmetrically. Measured W_2^2 distortion for $\mathcal{N}(0, 1)$:

m	equal-mass	Lloyd–Max	reduction	extreme reps (equal-mass $\rightarrow W_2$)
5	1.03×10^{-1}	7.99×10^{-2}	22%	
9	4.70×10^{-2}	2.79×10^{-2}	41%	$\pm 1.70 \rightarrow \pm 2.25$
15	2.42×10^{-2}	1.07×10^{-2}	56%	$\rightarrow \pm 2.68$
31	9.59×10^{-3}	2.70×10^{-3}	72%	$\rightarrow \pm 3.17$

Both decay as $W_2^2 \sim C/m^2$ (spike-discretization RMS error $\sim 1/m$); Lloyd–Max has the smaller constant, and the gap widens with m . In the current implementation the representatives are already W_2 -optimal (they are conditional means throughout), while default *edges* are equal-mass quantiles built once and reused; per-layer Lloyd–Max re-gridding of the exact mixture is available as a drop-in and only changes the edges. A practical caveat: past ≈ 4 –7 bins accuracy is limited by the bulk $K = 2$ closure, not the spike discretization, so W_2 -optimal binning mainly reaches the same floor with *fewer* bins (most useful in the few-bin / deeper-net regime).

8 Error sources and the width scaling law

8.1 Error budget

Three approximations enter:

1. **Spike discretization** — within-bin collapse of A ; equals $W_2^2(\mathcal{L}(A), \widehat{\mathcal{L}}(A))$ by (12); decays as $1/m^2$ and is minimized by Section 7. At least one negative bin is required (a single $(-\infty, 0)$ catch-bin suffices); dropping negatives entirely costs 30–60% error.
2. **Bulk $K = 2$ Gaussian closure** — B | bin treated as Gaussian. This is the dominant, width-dependent floor: relative MSE $\sim n^{-2}$, the budget rate.
3. **Mixture re-Gaussianization** — refitting the mixture (9) as one Gaussian per bin.

8.2 Measured scaling

A budget- $k_{\max}$ predictor has output error $O(n^{-k_{\max}})$, so at $K = 2$ the relative MSE $\sim n^{-2}$ (RMS $\sim n^{-1}$). Representing the spike coordinate with enough bins is exactly what recovers this rate; the naive $m = 1$ closure (collapse A to its mean and eat the ReLU Jensen gap) only reaches n^{-1} . Seed-averaged over random coordinate-spiked nets, depth 2, $m = 31$:

n	MSE (binned)	MSE ($m=1$)	RMS advantage
32	2.5×10^{-4}	9.7×10^{-2}	$19.5\times$
64	6.9×10^{-5}	4.2×10^{-2}	$24.8\times$
128	1.9×10^{-5}	1.7×10^{-2}	$30.2\times$
fit	MSE $\sim n^{-1.88}$ ($\approx n^{-2}$, the $K=2$ rate)	MSE $\sim n^{-1.25}$ ($\approx n^{-1}$, naive)	

Enough bins are needed to see the n^{-2} rate: with too few, the spike-discretization error floors the total and flattens the slope. The fixed-width refinement curve shows error dropping sharply with m and then converging to the bulk-closure floor; pushing *below* that floor requires higher bulk cumulant order (the $K > 2$ hook propagates conditional cumulant tensors per bin through ordinary harmonic kprop), not finer spike bins.

A Closed forms used

Standard normal $\varphi(x) = e^{-x^2/2}/\sqrt{2\pi}$, Φ its cdf. Truncated normal on $[a, b]$ with $Z = \Phi(b) - \Phi(a)$ (and $x\varphi(x) \rightarrow 0$ as $x \rightarrow \pm\infty$):

$$\mathbb{E}[A \mid a \leq A < b] = \frac{\varphi(a) - \varphi(b)}{Z}, \quad \text{Var} = 1 + \frac{a\varphi(a) - b\varphi(b)}{Z} - \left(\frac{\varphi(a) - \varphi(b)}{Z}\right)^2.$$

Rectified Gaussian, $Y \sim \mathcal{N}(\mu, \sigma^2)$, $z = \mu/\sigma$:

$$\mathbb{E}[\rho(Y)] = \mu\Phi(z) + \sigma\varphi(z), \quad \mathbb{E}[\rho(Y)^2] = (\mu^2 + \sigma^2)\Phi(z) + \mu\sigma\varphi(z), \quad P(Y \leq 0) = \Phi(-z).$$

B Implementation map

The $K = 2$ core (`Mecha_preds/binning_kprop/core.py`) is numpy/scipy and torch-free: `BinnedK2State`, `gaussian_initial_state`, `linear_step_k2`, `relu_step_k2`, `run_binned_kprop_k2`, `unconditional_mean[_cov]`; Gaussian-ReLU integrals come from the shared kernel `Mecha_preds/_utils`. Grid builders live in `binning.py` (`make_gaussian_edges`, `make_relu_post_edges`, `lloyd_max_edges[_mixture][_split]`). `kprop_hook.py` bridges to ordinary harmonic kprop for the "kprop" bulk-ReLU backend and the $K > 2$ per-bin tower. `adapter.py` provides the drop-in `run_binned_kprop(model, config={"num_bins": m})`; `selftest.py`, `scaling.py`, `benchmark.py` verify against Monte Carlo, check the n^{-2} law, and tune m respectively.